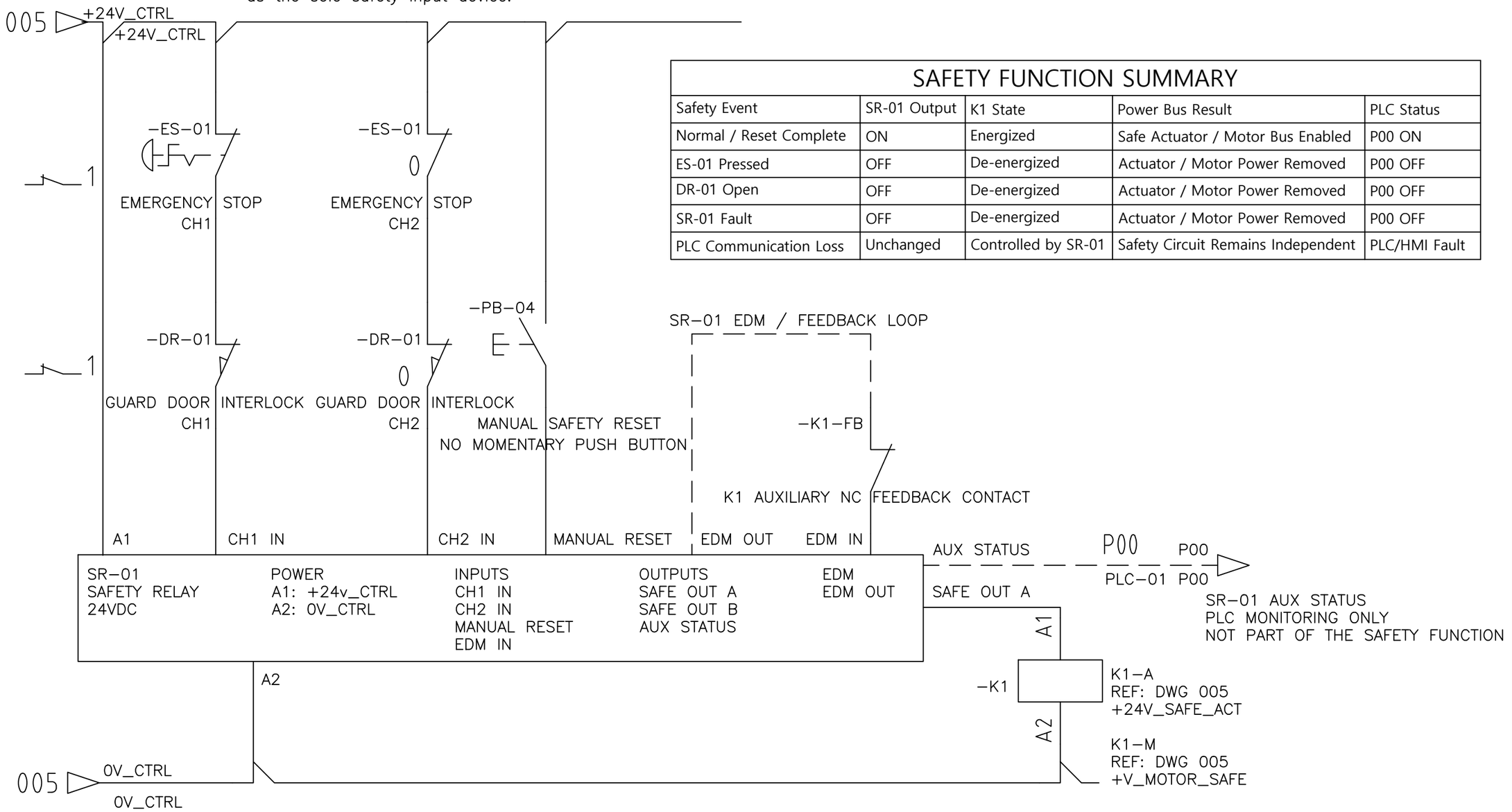
DR-01 shall be a safety-rated two-channel guard interlock.
A standard non-safety door sensor shall not be used
as the sole safety input device.



SAFETY FUNCTION SUMMARY

Safety Event	SR-01 Output	K1 State	Power Bus Result	PLC Status
Normal / Reset Complete	ON	Energized	Safe Actuator / Motor Bus Enabled	P00 ON
ES-01 Pressed	OFF	De-energized	Actuator / Motor Power Removed	P00 OFF
DR-01 Open	OFF	De-energized	Actuator / Motor Power Removed	P00 OFF
SR-01 Fault	OFF	De-energized	Actuator / Motor Power Removed	P00 OFF
PLC Communication Loss	Unchanged	Controlled by SR-01	Safety Circuit Remains Independent	PLC/HMI Fault

[GENERAL SAFETY DESIGN NOTES]

[1] ES-01 and DR-01 safety channels shall be hardwired directly to SR-01 and shall not depend on PLC logic.

[2] SR-01 shall use a monitored manual reset function. PB-04 Safety Reset shall be located outside the hazard area.

[3] PB-04 Safety Reset restores safety output permission only. It shall not initiate the automatic production cycle.

[4] K1-A and K1-M shall remove actuator and motor power when the SR-01 safety output is de-energized.

[5] PLC-01 safety status monitoring is diagnostic only and is not part of the safety function.

[6] Actual terminal assignments, safety category, PL/SIL rating, relay model, and contact ratings shall be verified using the selected safety device manufacturer's documentation before fabrication.

[7] The safety circuit shall be tested after installation by verifying E-Stop, door-open, manual-reset, and restart behavior.

NO.	DATE	REVISION
A	2026-08-24	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

Safety Circuit and Emergency Stop

DRAWN BY: J. H. KO
DATE: 2026-08-24

DWG NO: 006

SHEET NO: 1 OF 1